

Experiment 20: ECB vs CBC Error Propagation:

```
pt = input("Enter Plaintext: ")
# ECB Mode
ecb = [pt[i:i+4] for i in range(0, len(pt), 4)]
print("\nECB Blocks:", ecb)

# CBC Mode
iv = "INIT"
cbc = []
prev = iv

for block in ecb:
    x = block + "^" + prev
    cbc.append(x)
    prev = x

print("CBC Blocks:", cbc)

# Simulating error
print("\nIf error occurs in one ECB block:")
print("Only that block is affected")

print("\nIf error occurs in one CBC block:")
print("That block and next block are affected")
```

Output:

```
Enter Plaintext: hello

ECB Blocks: ['hell', 'o']
CBC Blocks: ['hell^INIT', 'o^hell^INIT']

If error occurs in one ECB block:
Only that block is affected

If error occurs in one CBC block:
That block and next block are affected

=== Code Execution Successful ===
```